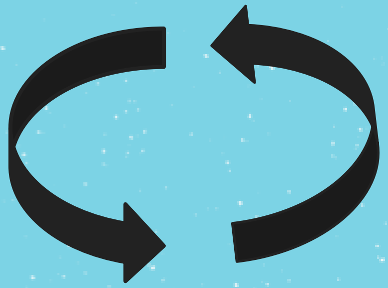
Topic 2
Motion Transmission and Transformation

MOTION AND FORCES

. For a technical object to **work properly** it often must be able to withstand certain **forces** (or **constraints**) and have a certain amount of **motion**

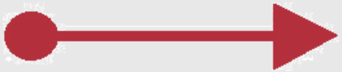
TYPES OF MOTION

Translational



Rotational

Helical

Types of motion	Unidirectional	Bidirectional
Translational		
Rotational		
Helical		

TRANMISSION vs TRANSFORMATION

- A **group of machines** can work together to make a **system.**
    - A system can **transmit or carry** the motion from one object to another

TRANMISSION vs TRANSFORMATION

A group of simple machines can also work together to make a **system** where the **type of motion is changed (transformed)** from one component to another

TRANMISSION VS TRANSFORMATION

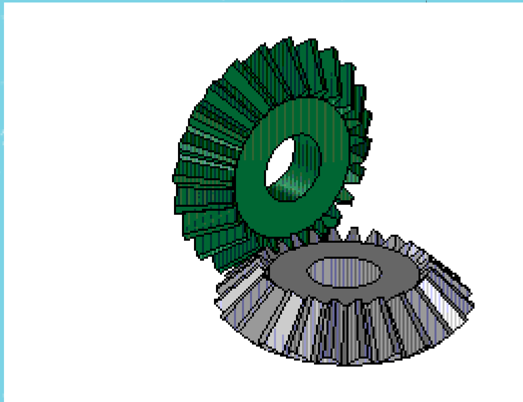
Transmission system:

- **Multiple components** that have the **same type** of movement (rotational, translational, or helical)

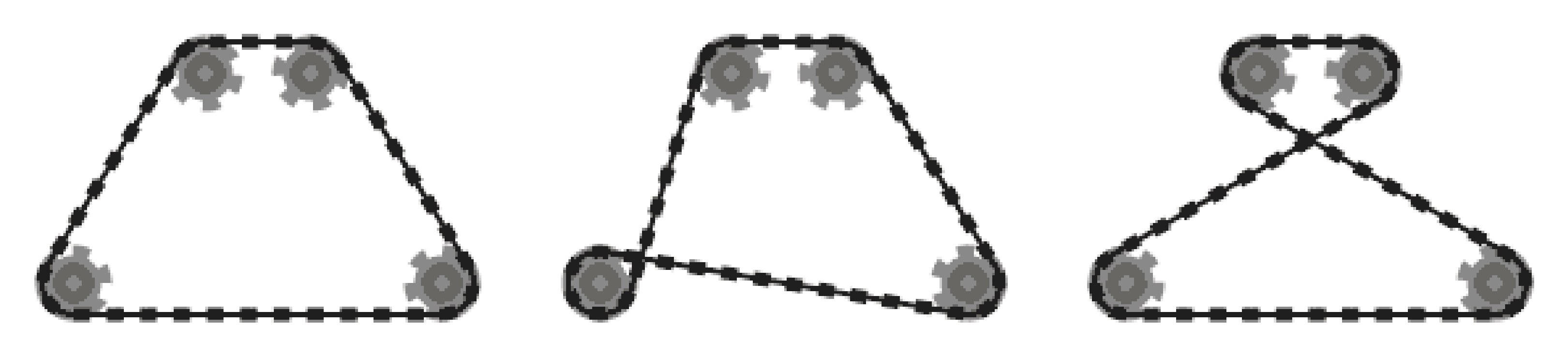
Transformation system:

- **Different components** in the system that have **different** types of movement
    - Ex: Rotational movement leads to translational movement

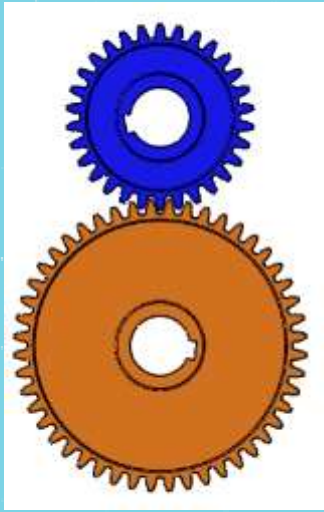
Components can be connected directly:



Or through an intermediate component:



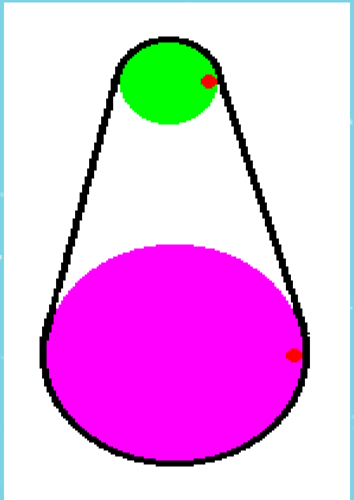
TRANSMISSION SYSTEMS



GEAR TRAINS

CHAIN & SPROCKET

FRICTION WHEELS

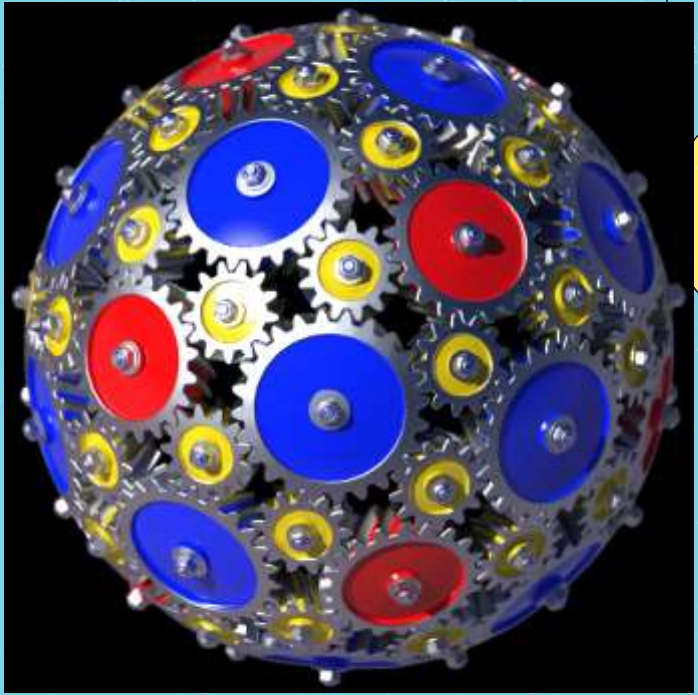


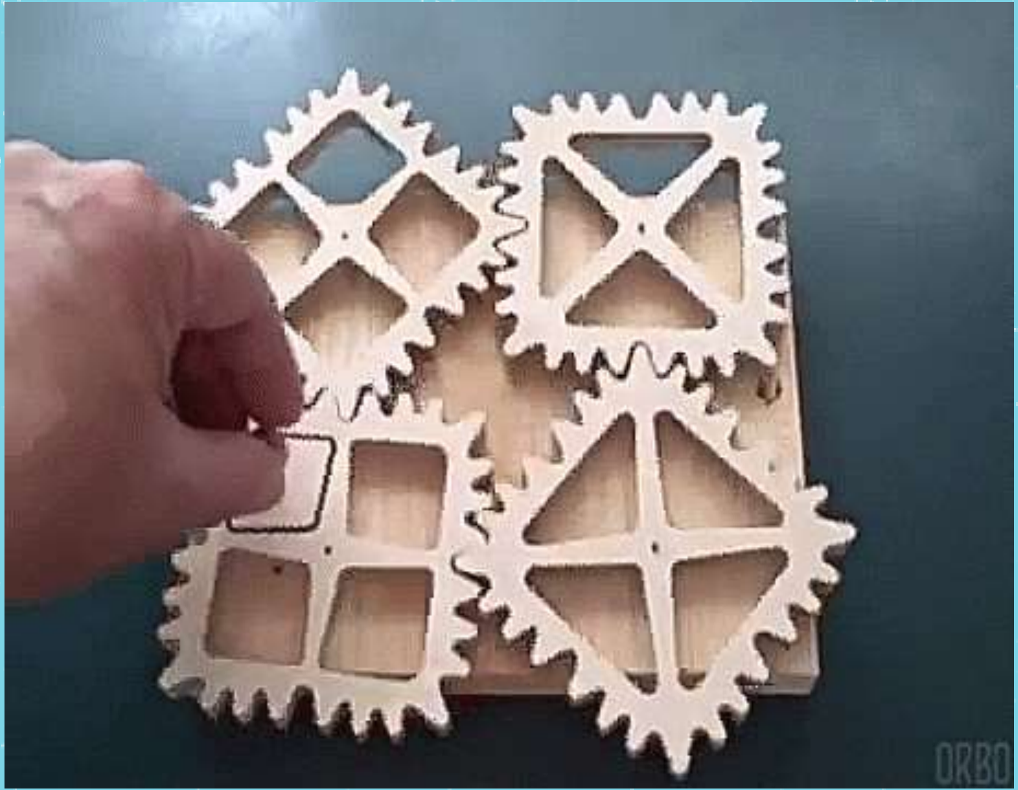
BELT & PULLEY

TRANSMISSION SYSTEMS

1) Gear Trains

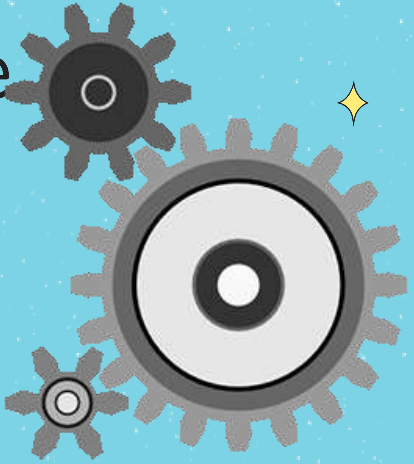
- At least **2 gears** whose teeth **interlock**




ORBO


TRANSMISSION SYSTEMS

- Notice that the gears move in **different** directions
- The **size** of the wheels (or **the number of teeth**) will affect the **speed of rotation**



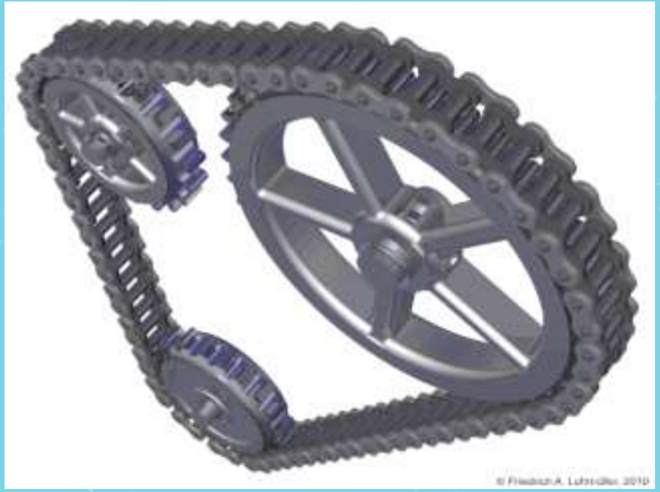
The small gear spins faster

TRANSMISSION SYSTEMS

2) Chain and Sprocket

- At least 2 gears connected by a chain

Notice: all the gears turn in the same direction

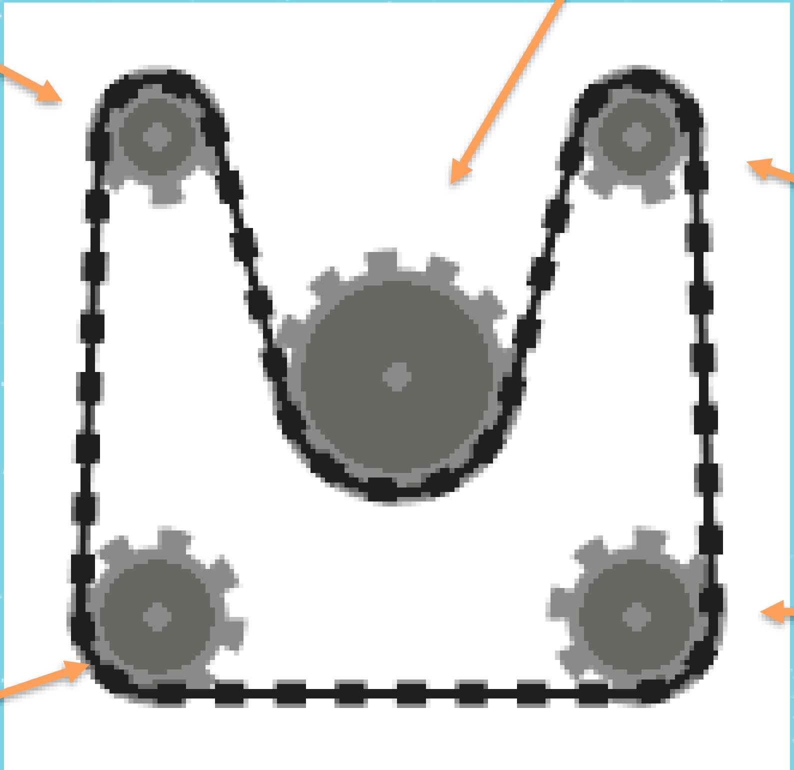


If it had gears on the **outside**; they would turn in the **opposite** direction

Chain & Sprocket

Inside Chain = Same direction

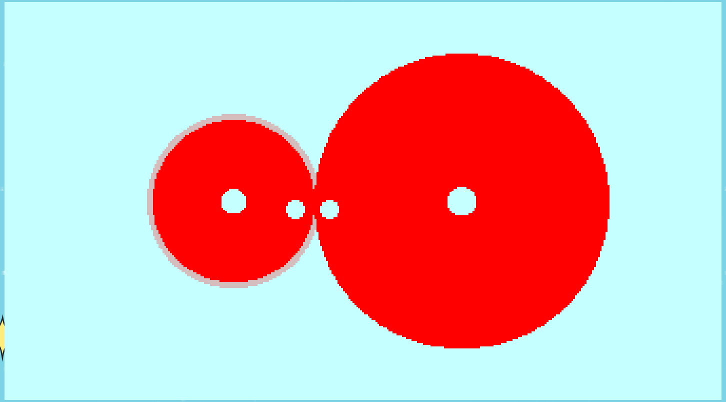
Outside Chain = Opposite direction



TRANSMISSION SYSTEMS

3) Friction Wheels (or friction gears)

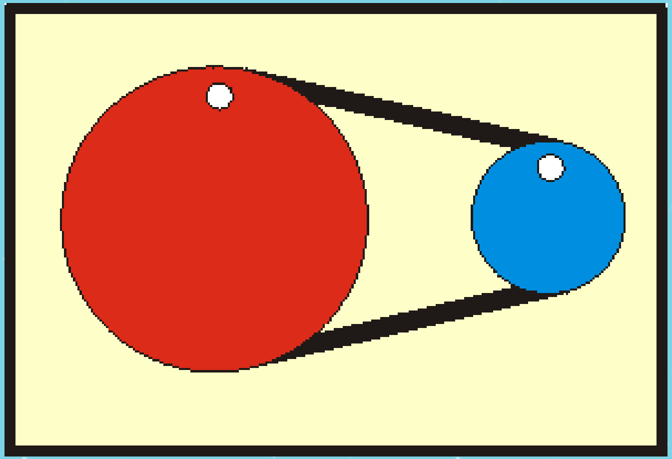
- Like gears but **without teeth**
- Touch **directly**; **friction** causes the movement



TRANSMISSION SYSTEMS

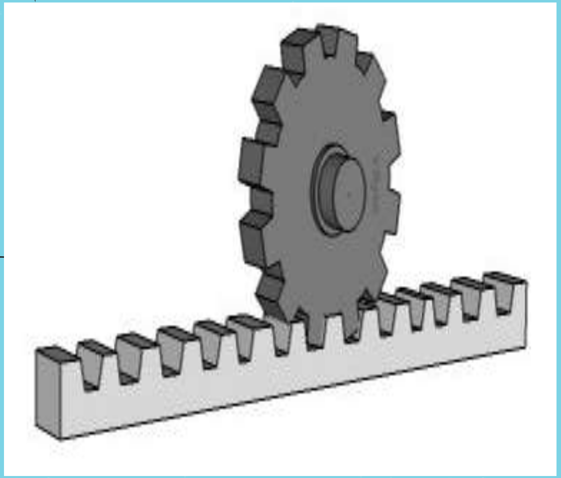
4) Belt and Pulley System

- Like chain and sprocket but **without teeth**

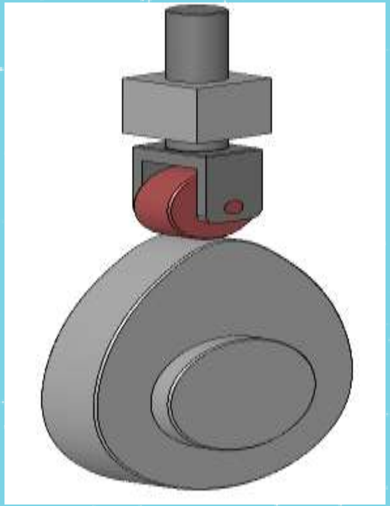


Without the teeth, there is a greater chance of the band slipping so it is not as efficient as the chain and sprocket

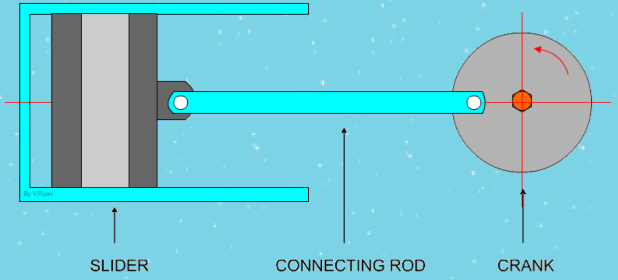
TRANSFORMATION SYSTEMS



RACK & PINION



CAM & FOLLOWER

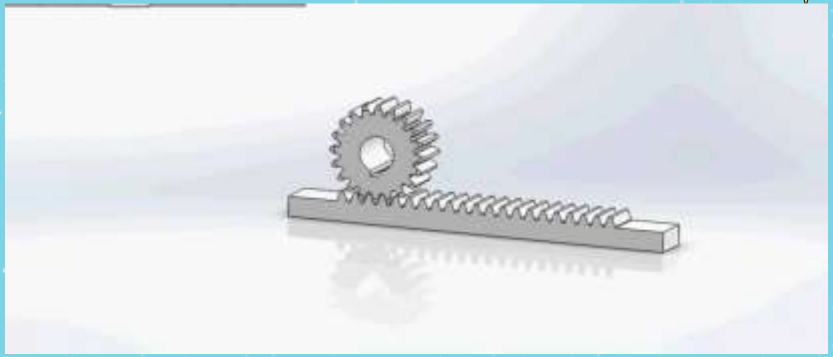


CONNECTING ROD & CRANK

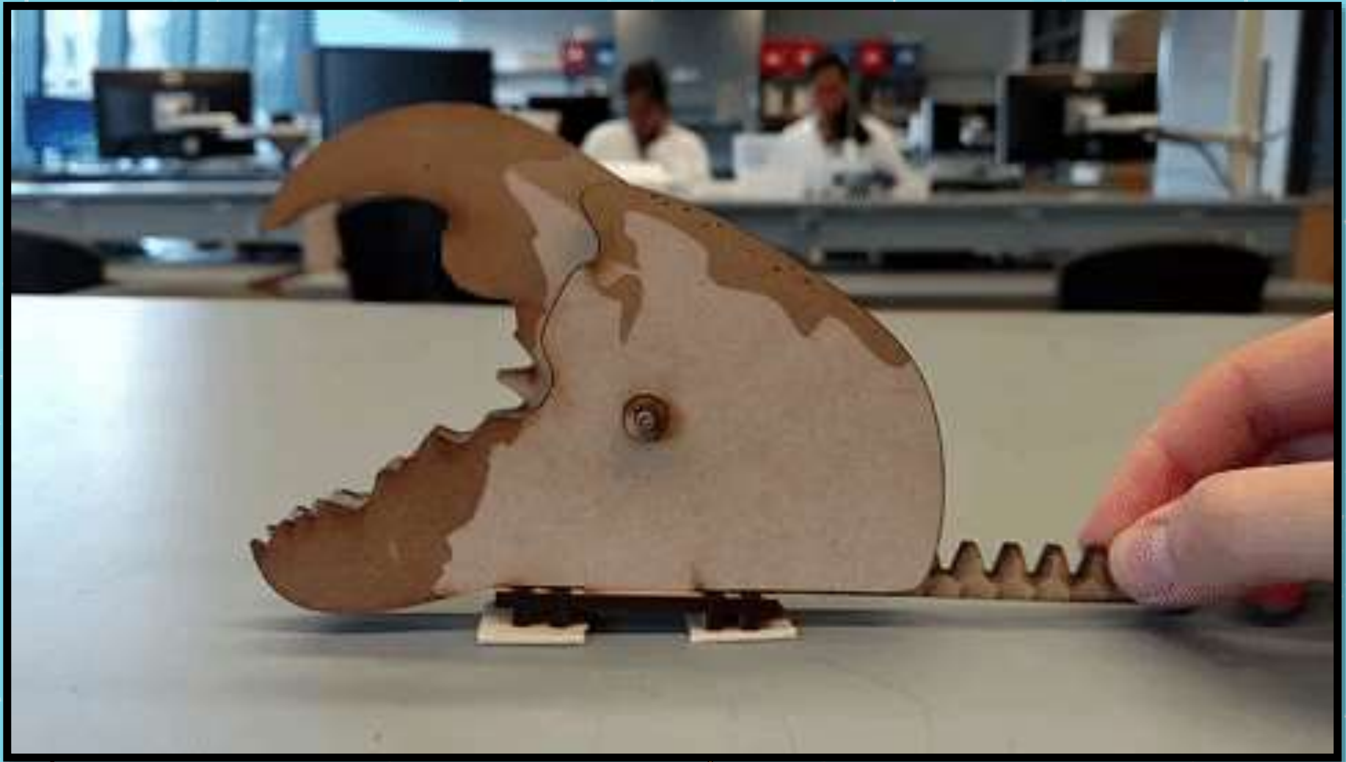
TRANSFORMATION SYSTEMS

1) Rack and Pinion

- Composed of a gear (**pinion**) and a grooved/toothed bar (**rack**)
- Transforms **rotational** into **translational motion** (and vice versa)



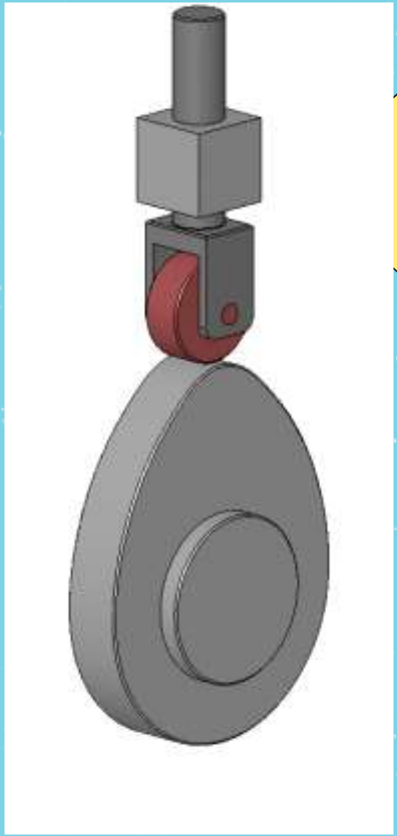
Rack and Pinion



TRANSFORMATION SYSTEMS

2) Cam and Follower

- Composed of a wheel that is often oval-shaped (**cam**) and a rod that raises and lowers (**follower**)
- Transforms rotational into translational motion (not reversible)

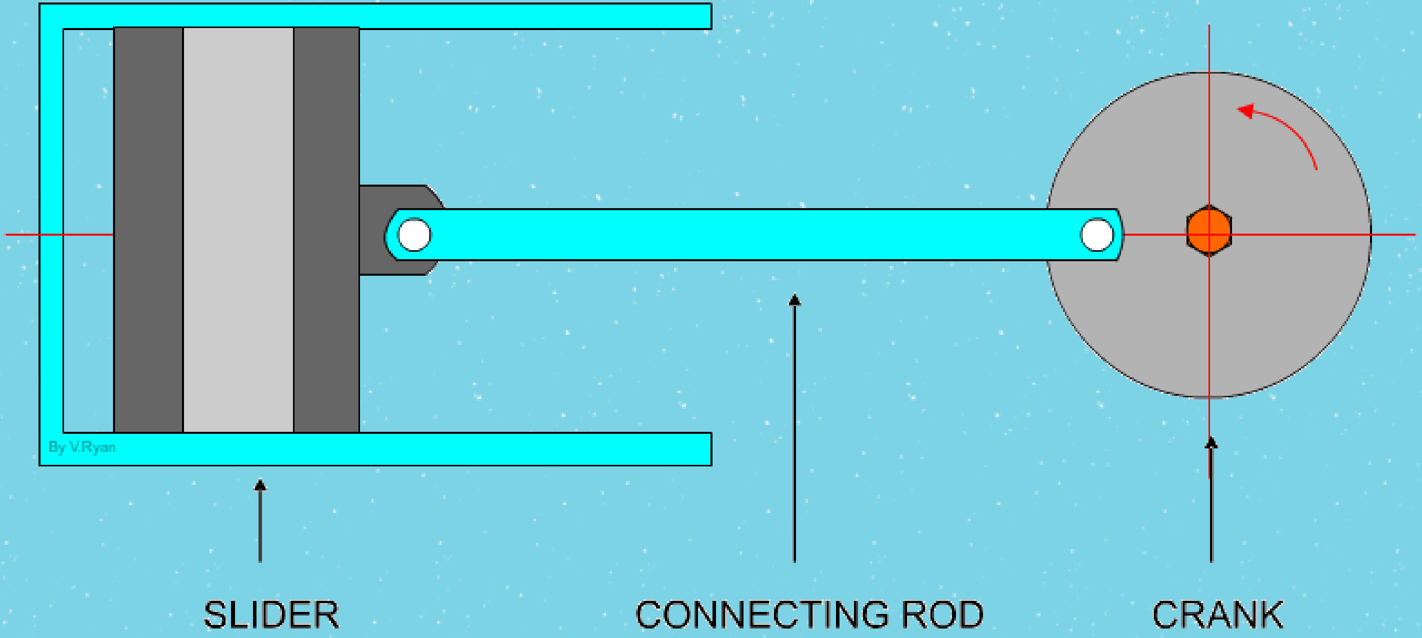


Cam and Follower

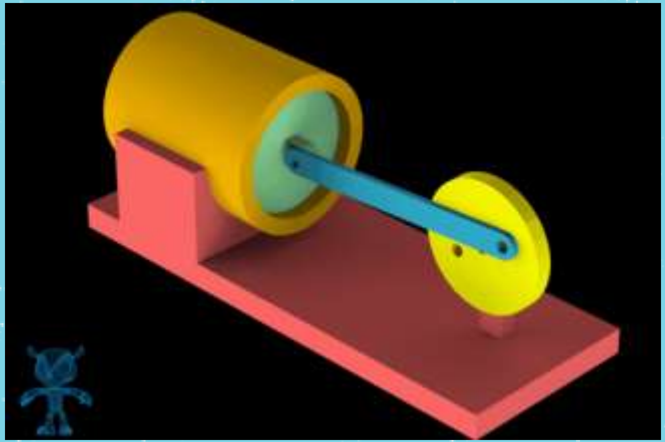
TRANSFORMATION SYSTEMS

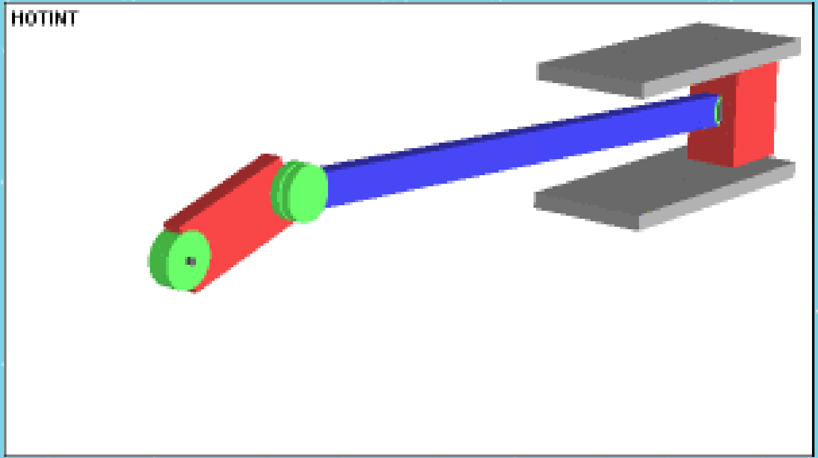
3) Connecting Rod and Crank (piston)

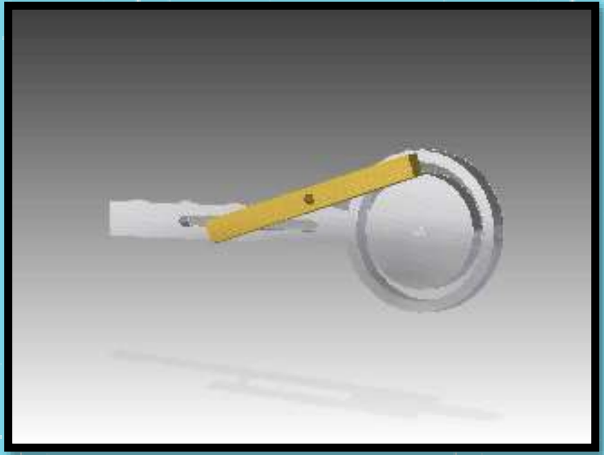
- Similar to cam and follower but the crank is usually a **circular wheel** and the slider is usually a **rod attached to a block that moves within the tube**


By V.Ryan
SLIDER
CONNECTING ROD
CRANK

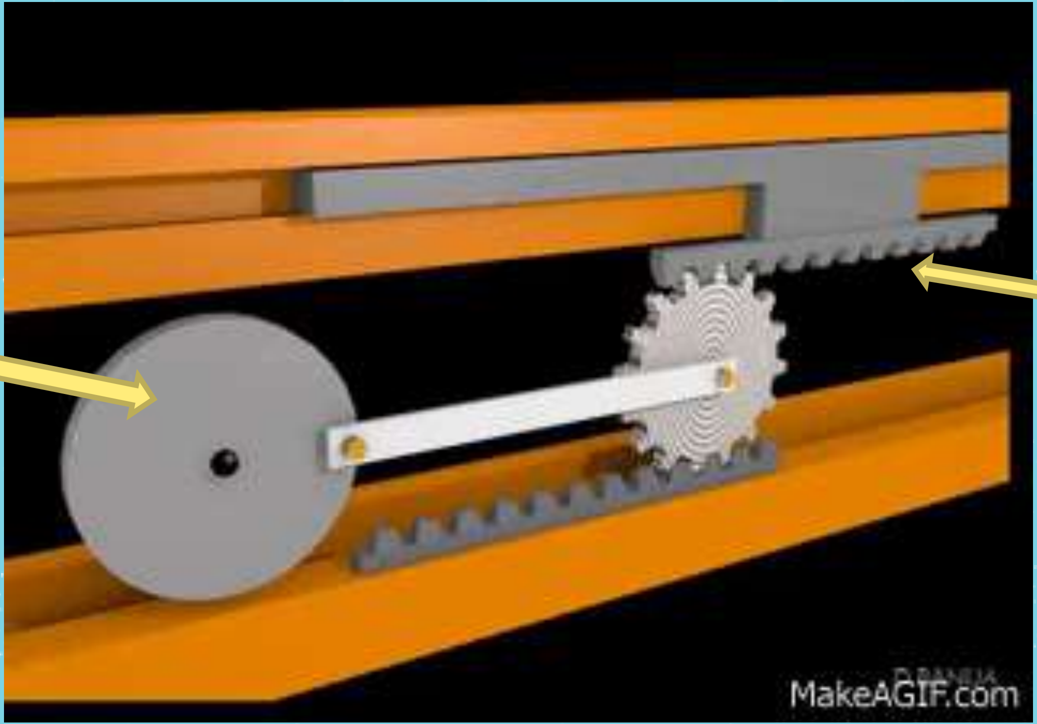
TRANSFORMATION SYSTEMS



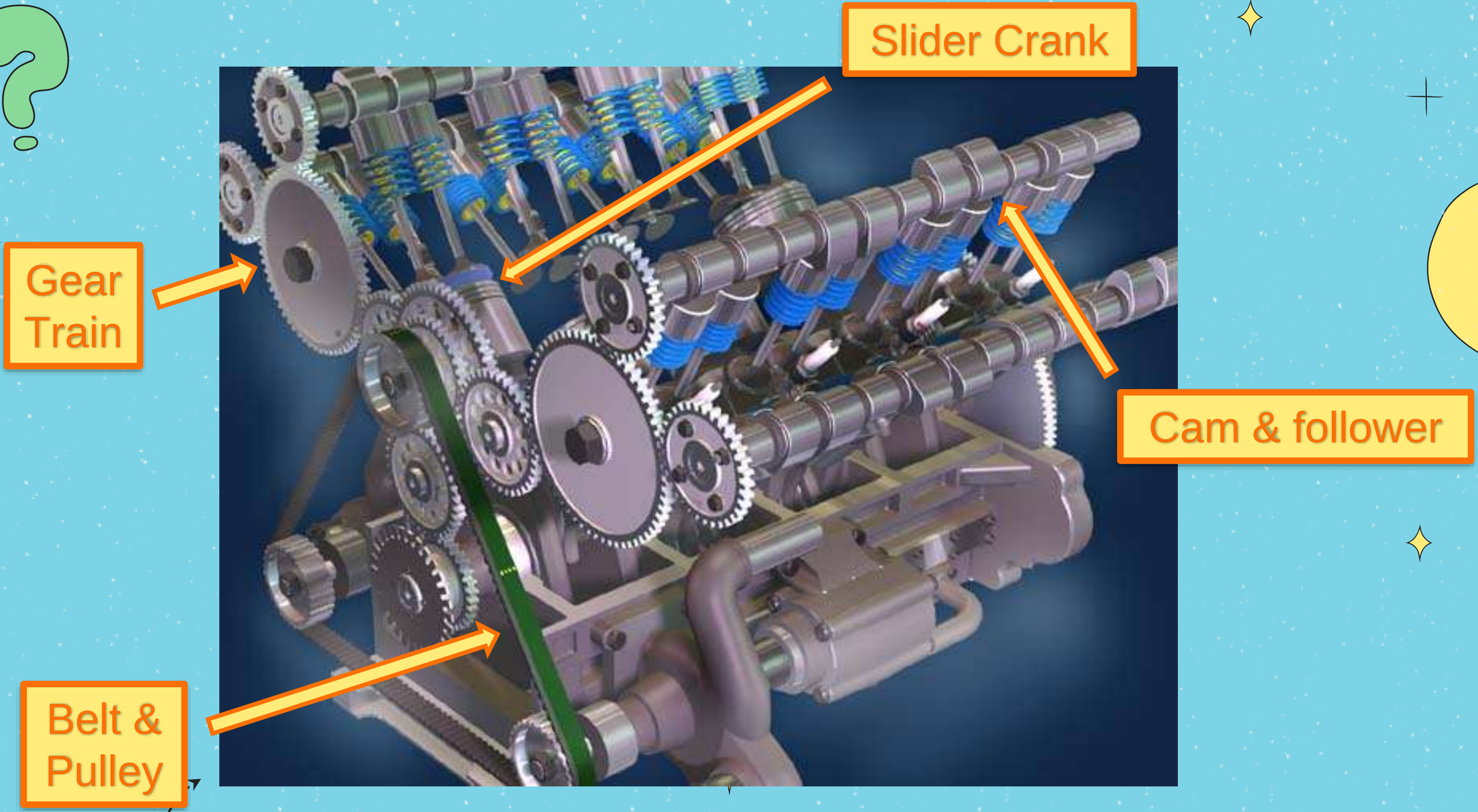




Slider & Crank



Rack & Pinion


Slider Crank
Gear Train
Cam & follower
Belt & Pulley